

# Paper VII — The BPB/MBE Acceleration-Scale Gate

BTFR Normalisation, Raw Baryonic Rotation Curves, and the Emergence of  $a_0$   
v0.3 — HAL corpus-aware P7-A7 manuscript lock

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## Abstract

Paper VII upgrades the BPB/MBE acceleration-scale test from the effective BTFR gate of v0.1 to the full P7-A1–P7-A7 source-locked sequence. The branch predicts an absolute acceleration scale without adding a galactic acceleration parameter,

$$a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}) = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}, \quad (1)$$

with  $H_0 = 68.910590682276 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $s_{\text{surv},0} = 0.7956924397708606$ . The same value is reproduced by the conditional D2 thermal bridge through  $T_{\text{eff}} = (1 - s_{\text{surv},0})T_{\text{dS}}$ . On the SPARC BTFR sample, the fixed- $a_0$  prediction gives  $n = 118$ ,  $\log_{10} M_b(100 \text{ km s}^{-1}) = 9.741034665243$ , median residual  $-0.056007711329$  dex, and RMS residual  $0.295494277101$  dex. A minimal intrinsic-scatter BTFR likelihood passes at the normalisation level: the free slope-4 offset is not required by BIC, while the free-slope model is preferred and is therefore kept as an explicit slope/interpolation guardrail. At raw point-level SPARC rotation-curve level, the fixed- $a_0$  RAR interpolation improves the baryon-only Newtonian curves by a fractional raw- $\chi^2$  improvement of  $0.911207525795$ , with median velocity bias  $0.026795035737$ . Under global  $Y_{\text{disk}}$ ,  $Y_{\text{bul}}$  and  $a_0$  profiling, the primary sample has best  $a_0/a_0^{\text{BPB}} = 0.807075252405$  and the fixed- $a_0$  penalty is  $0.051995539672$  of the fixed raw  $\chi^2$ . Paper VII v0.3 is therefore a fixed-acceleration BTFR/interpolation support paper. It is not a full galaxy-level SPARC rotation-curve likelihood, not a unique interpolation-function derivation, and not an unconditional action-level MOND limit.

## 1 Purpose of v0.3

Paper VII asks whether the strict BPB/MBE branch predicts a dimensional galactic acceleration scale. Version v0.1 answered this at the effective level: the branch scale  $a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0})$  was source-verified, the algebraic BTFR identity followed from  $g_N(r_t) = a_0^{\text{BPB}}$ , and the split-relic/Verlinde-like bridge reproduced the BTFR scaling under a holographic mode-counting axiom.

Version v0.3 keeps that acceleration-scale logic but adds the observational gates P7-A3–P7-A7. The paper is no longer only an intercept or transition-radius diagnostic. It now contains three layers:

1. an absolute  $a_0$  lock from the BPB survival descendant;
2. a fixed- $a_0$  SPARC BTFR normalisation and minimal likelihood test;
3. a raw point-level baryonic rotation-curve interpolation test with global nuisance profiling.

The central claim is deliberately bounded. BPB/MBE fixes a plausible acceleration scale and passes the BTFR/interpolation support layers. The full galaxy-level likelihood with distances, inclinations, stellar mass-to-light ratios, covariance, external-field effects and interpolation-family freedom remains a later gate. The comparison point is the standard MOND/RAR and SPARC rotation-curve phenomenology, but the claim made here is narrower: the BPB/MBE dictionary fixes the acceleration normalisation and tests its BTFR and baryonic-RC consequences, rather than deriving a final MOND interpolation family [1, 2, 3].

## 2 Corpus placement and release navigation

Paper VII is the local-acceleration descendant of the BPB/MBE corpus. It should be read after Paper 0, which defines the survival factor  $s_{\text{surv},0}$ , and alongside Paper II, which uses SPARC rotation curves for the galaxy-scale Weyl-kernel test. In the release package, the top-level navigation files are:

- 00\_README\_CORPUS: human entry point and quick orientation;
- 00\_MANIFEST\_CORPUS.csv: paper-by-paper manifest;
- 00\_CLAIMS\_INDEX.csv: claim-to-source-lock index;
- 00\_REVIEWER\_ROUTE: short audit route for reviewers.

The present paper contributes the Paper-VII rows of that corpus index. Its allowed claim is fixed- $a_0$  BTFR and baryonic-interpolation support. Its forbidden claims remain full SPARC rotation-curve likelihood closure, a unique interpolation-function derivation, active38 as an independent  $a_0$  anchor, and an unconditional action-level MOND limit.

## 3 P7 source-lock dashboard

Table 1 summarises the v0.3 gate chain. Figure 1 shows the same structure visually.

Table 1: Paper VII v0.3 gate dashboard.

| Gate          | Status                            | Locked role                                                                                                                                              |
|---------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| P7-A1         | PASS                              | Numeric acceleration lock: $a_0^{\text{BPB}} = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$ , $s_{\text{surv},0} = 0.795692439771$ .                 |
| P7-A2D2/A2D2b | PASS                              | Conditional D2 thermal bridge and provenance: $T_{\text{eff}} = 5.546676980150 \times 10^{-31} \text{ K}$ and $a_0(T_{\text{eff}}) = a_0^{\text{BPB}}$ . |
| P7-A3         | PASS                              | Fixed- $a_0$ BTFR normalisation: $n = 118$ , median residual $-0.056007711329 \text{ dex}$ , RMS $0.295494277101 \text{ dex}$ .                          |
| P7-A4         | PASS with slope guardrail         | Minimal BTFR likelihood: $\sigma_{\text{int}} = 0.295494277101 \text{ dex}$ ; free slope-4 offset not required by BIC; free-slope model preferred.       |
| P7-A5         | PASS with raw- $\chi^2$ guardrail | Raw baryonic RC interpolation: fractional improvement $0.911207525795$ , raw reduced $\chi^2 = 56.310851078207$ .                                        |
| P7-A6/A6b     | PASS with full-RC guardrail       | Global nuisance profile: primary best $a_0/a_0^{\text{BPB}} = 0.807075252405$ ; active38 kept as shape-only cross-check.                                 |
| P7-A7         | LOCKED                            | Manuscript lock: fixed- $a_0$ BTFR/interpolation support, not full SPARC RC likelihood.                                                                  |

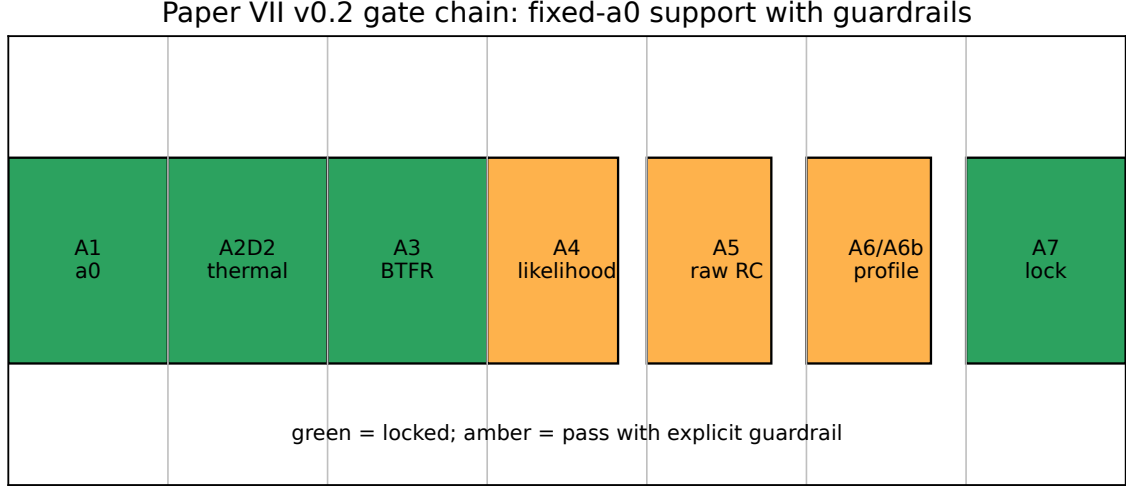


Figure 1: P7-A1–P7-A7 gate chain. Green indicates a direct lock. Amber indicates a pass with an explicit guardrail.

## 4 Acceleration-scale lock

The v0.3 numerical lock uses the exact P7-A7 values

$$H_0 = 68.910590682276 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (2)$$

$$s_{\text{surv},0} = 0.7956924397708606, \quad (3)$$

$$1 - s_{\text{surv},0} = 0.2043075602291394. \quad (4)$$

With  $c = 299792458 \text{ m s}^{-1}$  and  $1 \text{ Mpc} = 3.0856775814913673 \times 10^{22} \text{ m}$ , the acceleration descendant is

$$a_0^{\text{BPB}} = cH_0(1 - s_{\text{surv},0}) = 1.3678565925951916 \times 10^{-10} \text{ m s}^{-2}. \quad (5)$$

Relative to the diagnostic reference  $a_0^{\text{MOND}} = 1.2 \times 10^{-10} \text{ m s}^{-2}$ ,

$$\frac{a_0^{\text{BPB}}}{a_0^{\text{MOND}}} = 1.139880493829, \quad (6)$$

so the BPB value is +13.988049% above that reference. Figure 2 shows the comparison.

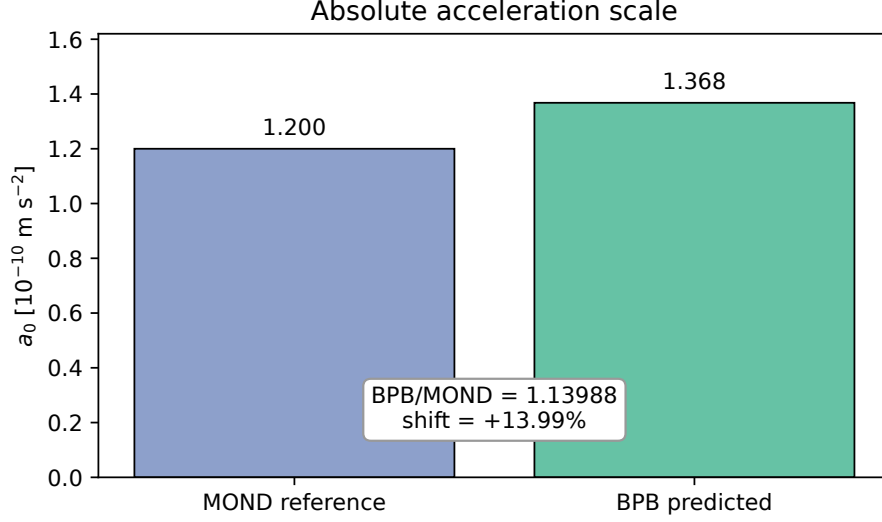


Figure 2: Absolute acceleration scale. BPB predicts  $a_0^{\text{BPB}} = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$ , about 13.99% above the conventional MOND reference value used here only as a diagnostic ruler.

The relation is not fitted to SPARC. The scale is inherited from the same survival factor used elsewhere in the BPB/MBE dictionary. The Paper-II D7 CMB-background cross-check changes  $a_0$  only by  $-0.224261\%$ , so the present value is stable at the sub-percent level under the current Hubble-anchor choice.

## 5 Conditional D2 thermal bridge

The thermal bridge rewrites the same identity using the de Sitter temperature

$$T_{\text{dS}} = \frac{\hbar H_0}{2\pi k_B}, \quad (7)$$

and the survival-weighted effective temperature

$$T_{\text{eff}} = (1 - s_{\text{surv},0})T_{\text{dS}}. \quad (8)$$

Applying the Unruh conversion  $a = 2\pi c k_B T / \hbar$  gives

$$\frac{2\pi c k_B T_{\text{eff}}}{\hbar} = c H_0 (1 - s_{\text{surv},0}) = a_0^{\text{BPB}}. \quad (9)$$

Numerically,

$$T_{\text{dS}} = 2.714866240842426 \times 10^{-30} \text{ K}, \quad (10)$$

$$T_{\text{eff}} = 5.546676980149714 \times 10^{-31} \text{ K}, \quad (11)$$

$$a_0(T_{\text{eff}}) = 1.3678565925951914 \times 10^{-10} \text{ m s}^{-2}. \quad (12)$$

The residual relative to the P7-A1  $a_0$  value is  $-1.89 \times 10^{-16}$  in relative units. This is an exact numerical identity at displayed precision.

The guardrail is important. P7-A2D2 is source-locked to the Paper-0 D2 conditional H1–H4 provenance bundle. It is a conditional thermal route to the same acceleration descendant, not an unconditional action-level derivation of MOND and not a BTFR likelihood.

## 6 Effective BTFR identity

Let  $r_t$  denote an effective transition radius at which the Newtonian baryonic acceleration reaches the BPB scale:

$$g_N(r_t) = \frac{GM_b}{r_t^2} = a_0^{\text{BPB}}. \quad (13)$$

Then

$$r_t^2 = \frac{GM_b}{a_0^{\text{BPB}}}. \quad (14)$$

At that radius,

$$V_f^2 \simeq g_N(r_t)r_t = a_0^{\text{BPB}}r_t. \quad (15)$$

Squaring gives

$$V_f^4 = (a_0^{\text{BPB}})^2 r_t^2 = GM_b a_0^{\text{BPB}}. \quad (16)$$

Thus the BTFR slope follows once the effective transition condition is accepted. The open theoretical problem is still the microscopic selection of  $r_t$  from the bare BPB/MBE field equations. The v0.3 paper does not hide this; it tests the fixed scale against SPARC observables while keeping the radius-selection problem open.

## 7 SPARC BTFR normalisation

P7-A3 tests the fixed acceleration scale against the SPARC baryonic Tully–Fisher relation using the SPARC baryonic catalogue and flat-velocity convention [2]:

$$M_b^{\text{pred}} = \frac{V_{\text{flat}}^4}{Ga_0^{\text{BPB}}}. \quad (17)$$

The baryonic convention is fixed, not profiled, at this stage:

$$M_b = 0.500 L_{3.6} + 1.330 M_{\text{HI}}. \quad (18)$$

The predicted zero point at  $V_{\text{flat}} = 100 \text{ km s}^{-1}$  is

$$\log_{10} M_b(100 \text{ km s}^{-1}) = 9.741034665243. \quad (19)$$

For the primary finite SPARC sample, P7-A3 gives the summary in Table 2 and Figure 3.

Table 2: P7-A3 fixed- $a_0$  BTFR normalisation. Residuals are  $\log_{10}(M_{\text{obs}}/M_{\text{BPB}})$ .

| Sample                 | $n$ | median [dex]    | RMS [dex]      | free slope     | median $a_{0,\text{inf}}/a_0^{\text{BPB}}$ |
|------------------------|-----|-----------------|----------------|----------------|--------------------------------------------|
| Primary finite         | 118 | −0.056007711329 | 0.295494277101 | 3.374907630412 | 1.137746644983                             |
| Quality 1              | 82  | −0.100657954938 | 0.266175008482 | 3.528260316132 | 1.260834974396                             |
| Paper-II active38      | 38  | −0.082524105026 | 0.219939973556 | 3.448644440822 | 1.210386025351                             |
| $V_{\text{flat}} > 50$ | 113 | −0.071518613455 | 0.255460308488 | 3.542563225344 | 1.179013053556                             |

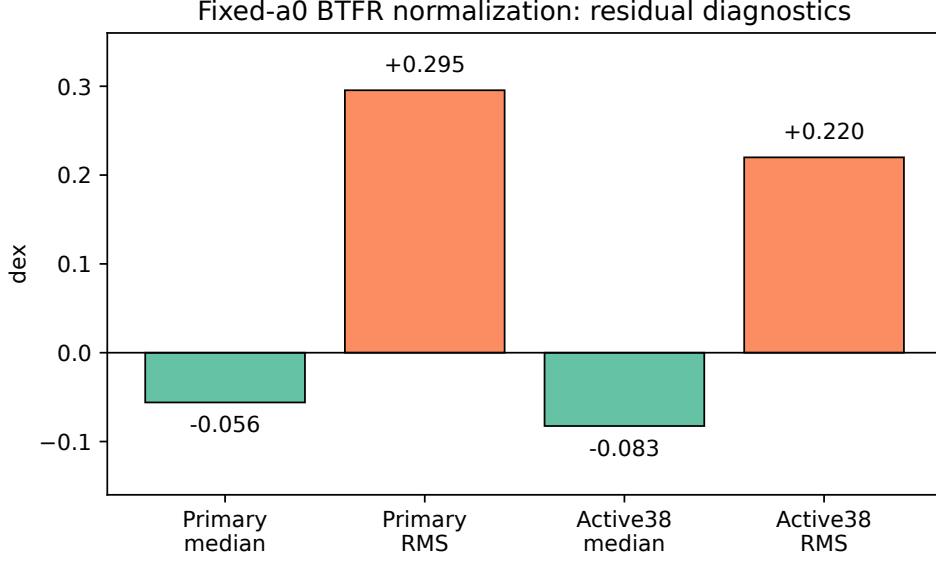


Figure 3: BTFR residual diagnostics for the fixed BPB acceleration scale. The primary median residual is  $-0.0560$  dex and the RMS scatter is  $0.2955$  dex.

A mass-to-light sensitivity scan shows that  $Y_{3.6} = 0.6$  nearly centers the BTFR normalisation, giving median residual  $-0.010880$  dex and fixed-slope-4 offset  $-0.001155$  dex. The v0.3 lock nevertheless keeps the primary convention at  $Y_{3.6} = 0.5$  so that the fixed- $a_0$  claim is not made by tuning the stellar mass-to-light ratio.

## 8 Minimal BTFR likelihood

P7-A4 promotes the normalisation test to a minimal Gaussian intrinsic-scatter likelihood. The BPB model fixes the slope to 4, fixes the zero point through  $a_0^{\text{BPB}}$ , and profiles only the intrinsic scatter. It is compared with two alternatives: a free slope-4 intercept and a free intercept plus free slope. Table 3 gives the primary sample result.

Table 3: P7-A4 primary BTFR likelihood models.  $\Delta\text{AIC}$  and  $\Delta\text{BIC}$  are relative to the fixed BPB model.

| Model                      | slope          | $\alpha - \alpha_{\text{BPB}}$ | $\sigma_{\text{int}}$ [dex] | $\Delta\text{AIC}$ | $\Delta\text{BIC}$ |
|----------------------------|----------------|--------------------------------|-----------------------------|--------------------|--------------------|
| BPB fixed $a_0$ , slope 4  | 4.000000       | 0                              | 0.295494277101              | 0                  | 0                  |
| Free offset, slope 4       | 4.000000       | $-0.047842362919$              | 0.291595569428              | $-1.134472660076$  | $+1.636211964390$  |
| Free intercept, free slope | 3.374907630412 | $+0.004283392318$              | 0.254811027061              | $-30.958035659962$ | $-25.416666411030$ |

The interpretation is precise. A free slope-4 offset is not required by BIC, so the fixed BPB normalisation survives the minimal likelihood gate. But the free-slope model is preferred, with slope 3.374907630412 and  $\Delta\text{BIC} = -25.416666411030$  relative to the fixed BPB model. This is the slope/interpolation guardrail, not a failure of the fixed- $a_0$  normalisation layer. Figure 4 visualises the BIC comparison.

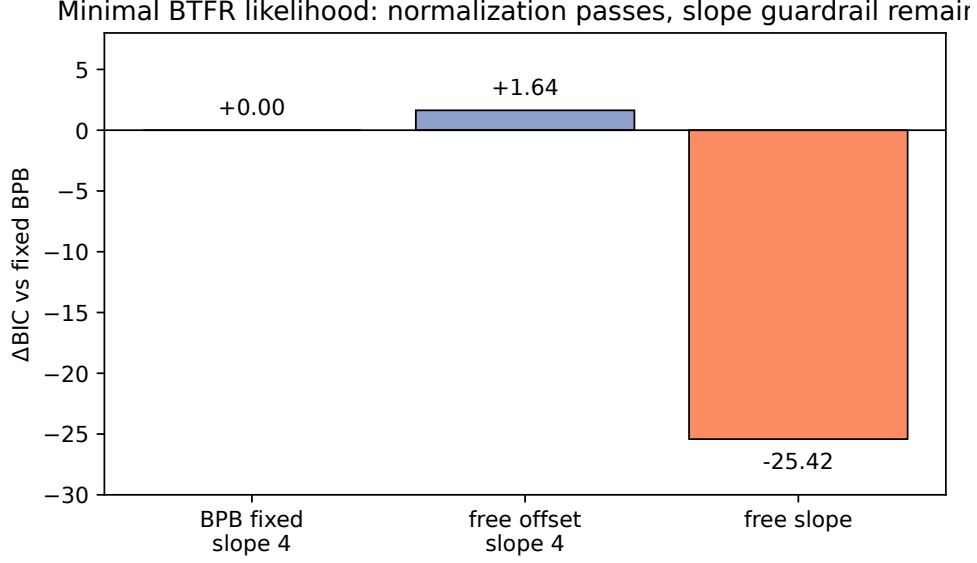


Figure 4: Minimal BTFR likelihood. A free slope-4 offset is not required under BIC, while the free-slope model is preferred. The v0.3 claim is therefore fixed- $a_0$  normalisation support, not full slope closure.

## 9 Raw point-level baryonic rotation curves

P7-A5 injects the fixed BPB acceleration scale into point-level SPARC baryonic rotation curves. The primary interpolation is the empirical RAR form used here as a diagnostic baryonic-interpolation law [3]

$$g = \frac{g_{\text{bar}}}{1 - \exp[-\sqrt{g_{\text{bar}}/a_0^{\text{BPB}}}]}, \quad (20)$$

with fixed  $Y_{\text{disk}} = 0.500$  and  $Y_{\text{bul}} = 0.700$  for this gate.

For the P7-A3 primary galaxy set, the baryon-only Newtonian raw  $\chi^2$  is

$$\chi_{\text{baryon}}^2 = 1,751,618.840117, \quad (21)$$

whereas the fixed- $a_0$  RAR interpolation gives

$$\chi_{\text{RAR}}^2 = 155,530.570678. \quad (22)$$

The fractional improvement is

$$\frac{\chi_{\text{baryon}}^2 - \chi_{\text{RAR}}^2}{\chi_{\text{baryon}}^2} = 0.911207525795. \quad (23)$$

The median velocity bias is

$$\text{median} \left( \frac{V_{\text{pred}} - V_{\text{obs}}}{V_{\text{obs}}} \right) = 0.026795035737, \quad (24)$$

and the RMS velocity residual is  $23.529397 \text{ km s}^{-1}$ . Figure 5 shows the raw- $\chi^2$  change.

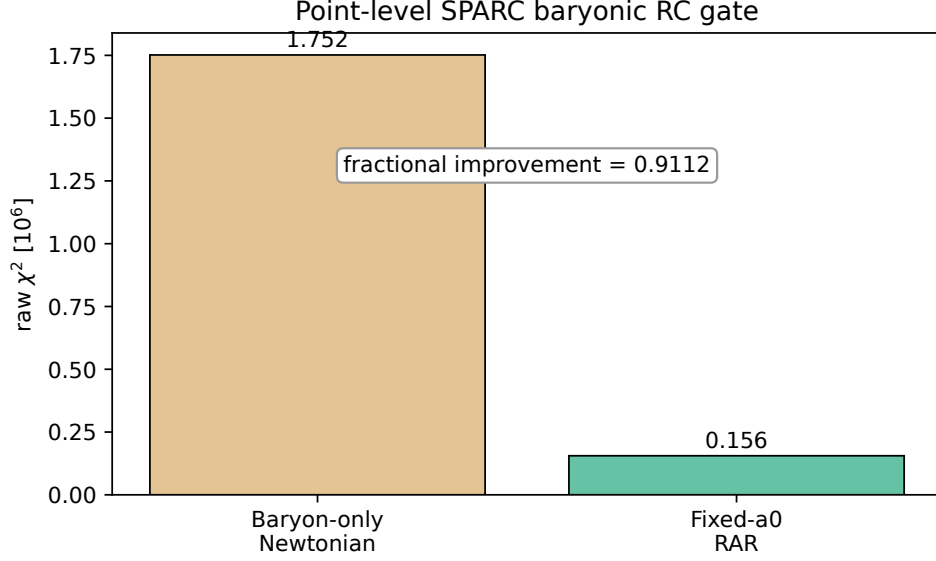


Figure 5: Raw point-level SPARC baryonic RC gate. The fixed- $a_0$  RAR interpolation strongly improves the baryon-only Newtonian prediction, but the raw reduced  $\chi^2$  remains high.

Table 4 records the main P7-A5 sample summary.

Table 4: P7-A5 raw baryonic RC interpolation summary.

| Sample/model              | points | galaxies | raw $\chi^2$     | reduced raw $\chi^2$ | fractional improvement |
|---------------------------|--------|----------|------------------|----------------------|------------------------|
| Primary baryon-only       | 2762   | 118      | 1,751,618.840117 | 634.184953           | 0                      |
| Primary fixed- $a_0$ RAR  | 2762   | 118      | 155,530.570678   | 56.310851            | 0.911207525795         |
| Active38 fixed- $a_0$ RAR | 714    | 38       | 27,949.675891    | 39.145204            | 0.894336858780         |
| All MRT fixed- $a_0$ RAR  | 3389   | 175      | 176,956.306455   | 52.214903            | 0.909737593470         |

The guardrail is also numerical: the primary raw reduced  $\chi^2$  remains 56.310851078207. Therefore P7-A5 is a raw-shape support gate, not a full SPARC rotation-curve likelihood.

## 10 Global nuisance profile and active38 boundary audit

P7-A6 profiles global  $Y_{\text{disk}}$ ,  $Y_{\text{bul}}$  and  $a_0/a_0^{\text{BPB}}$  on the raw SPARC points. P7-A6b then extends the active38 scan below the previous  $a_0/a_0^{\text{BPB}} = 0.55$  lower boundary.

For the primary sample, the extended-grid best is

$$a_0/a_0^{\text{BPB}} = 0.807075252405, \quad (25)$$

$$Y_{\text{disk}} = 0.525, \quad (26)$$

$$Y_{\text{bul}} = 0.800, \quad (27)$$

$$\chi_{\text{best}}^2 = 147,443.674720. \quad (28)$$

The fixed-minus-best raw- $\chi^2$  penalty is

$$\frac{\chi_{\text{fixed}}^2 - \chi_{\text{best}}^2}{\chi_{\text{fixed}}^2} = 0.051995539672. \quad (29)$$

Thus the fixed BPB acceleration scale remains close at the global nuisance-profiled level.



For the Paper-II active38 subset, the extended-grid best is

$$a_0/a_0^{\text{BPB}} = 0.486722101981, \quad (30)$$

$$Y_{\text{disk}} = 0.725, \quad (31)$$

$$Y_{\text{bul}} = 0.600. \quad (32)$$

This best point is not at the lower or upper scan boundary. But active38 is not promoted to an independent  $a_0$  anchor. It remains a shape/interpolation cross-check because it is a deliberately selected active subset from Paper II.

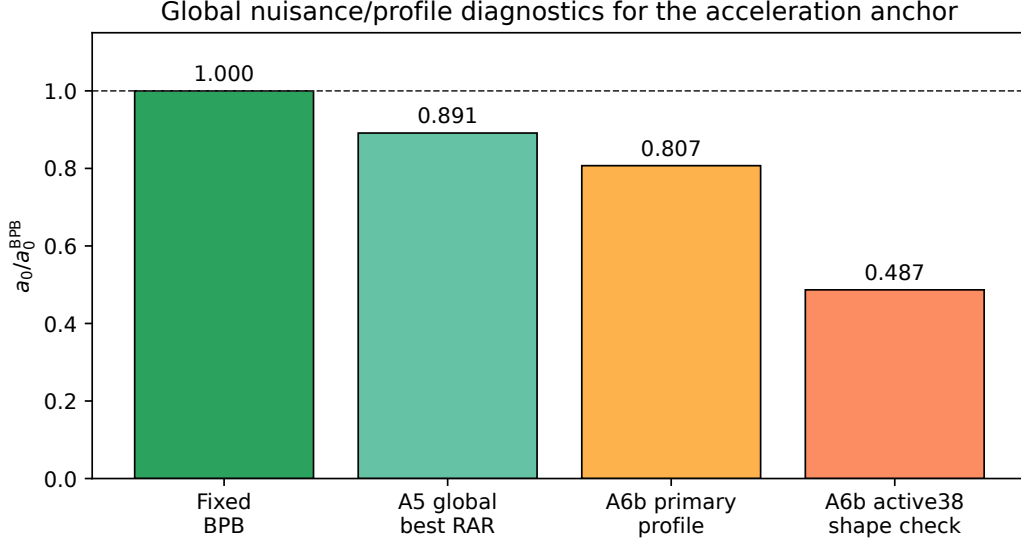


Figure 6: Acceleration-scale profile diagnostics. The primary sample remains within about 20% of the fixed BPB value under global nuisance profiling. The active38 optimum is lower and is retained only as a shape cross-check.

## 11 Split-relic force-law bridge

The v0.1 split-relic/Verlinde-like bridge remains useful as a theoretical diagnostic. It gives the effective force law

$$g_{\text{tot}} = g_N + \sqrt{a_0^{\text{BPB}} g_N}. \quad (33)$$

For  $g_N \ll a_0^{\text{BPB}}$ , the square-root term dominates,

$$g_{\text{tot}} \simeq \sqrt{a_0^{\text{BPB}} g_N}, \quad (34)$$

and circular motion again gives

$$V_f^4 = GM_b a_0^{\text{BPB}}. \quad (35)$$

The source diagnostic reports a benchmark agreement of 0.071% under the holographic mode-counting axiom  $N_{\text{modes}} = S_{\text{dS}}/k_B$ .

This bridge is not promoted beyond its source status. It is quasi-closed under the external holographic axiom, not derived from the bare BPB/MBE action alone. The two visually similar numbers from v0.1 remain distinct: the 0.071% force-law benchmark residual and the old  $-0.071$  dex intercept offset are different observables with different units.

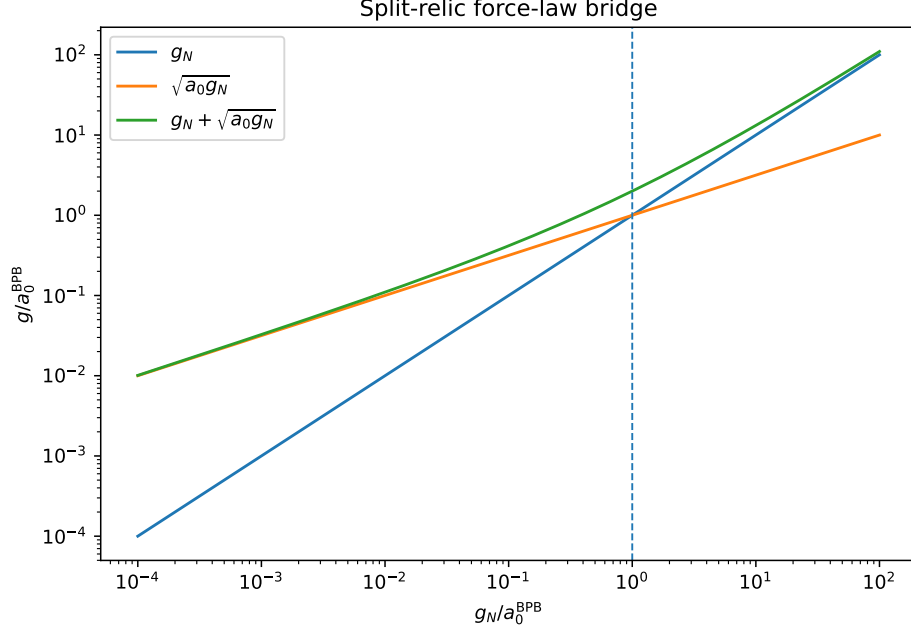


Figure 7: Split-relic force-law bridge retained from v0.1. At low acceleration, the square-root term produces the BTFR scaling.

## 12 Corpus synthesis and beta-closure check

The acceleration-scale gate is not a standalone phenomenological add-on. It uses the same survival descendant that appears throughout the BPB/MBE dictionary. The corpus-level reading is therefore:

Eight papers, one split-relic dictionary, with distinct observable projections.

The important distinction is that Paper VII tests a local acceleration-scale descendant, while Papers I–VI tested low-redshift growth, Weyl/lensing response, galaxy morphology kernels, CMB consistency, neutron-star tidal response, early-galaxy formation and the  $E_G$  Weyl–matter cross response.

The v0.1 cross-paper beta check remains valid. Paper IV introduced the neutron-star closure variable

$$\beta_{\text{cl}} = -0.12489553, \quad (36)$$

which is not the same object as the low-redshift growth value  $\beta_{\text{MAP}} = -0.2106621406$ . The closure check is

$$\alpha_{\infty} = \frac{|\beta_{\text{cl}}|}{s_{\text{surv},0}} = 0.1569645797, \quad (37)$$

matching the source candidate at the  $10^{-9}$  level. The correct wording is therefore  $\beta_{\text{cl}}$  or  $\beta_{\text{NS,closure}}$ , not simply  $\beta_{\text{grow}}$ .

## 13 What is closed and what remains open

### 13.1 Closed in v0.2

1. The acceleration scale is source-locked:  $a_0^{\text{BPB}} = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$ .
2. The D2 thermal bridge reproduces the same value through  $T_{\text{eff}} = (1 - s_{\text{surv},0})T_{\text{dS}}$ , with D2 provenance locked and conditional guardrail retained.
3. The effective BTFR identity  $V_f^4 = GM_b a_0^{\text{BPB}}$  follows from  $g_N(r_t) = a_0^{\text{BPB}}$ .

4. The fixed- $a_0$  SPARC BTFR normalisation passes: median residual  $-0.0560$  dex and RMS  $0.2955$  dex on  $n = 118$ .
5. The minimal BTFR likelihood passes at the fixed-normalisation level: a free slope-4 offset is not required by BIC.
6. The free-slope preference is recorded as a slope/interpolation guardrail.
7. The fixed- $a_0$  RAR interpolation strongly improves raw point-level SPARC baryonic rotation curves over baryon-only curves.
8. Global nuisance profiling keeps the primary sample's best  $a_0$  close to the BPB scale, with fixed penalty fraction  $0.051996$ .
9. Active38's lower optimum is resolved as non-boundary but remains shape-only.

### 13.2 Open after v0.2

1. The microscopic selection of  $r_t$  is not derived from the bare BPB/MBE equations.
2. The holographic mode-counting axiom behind the split-relic/Verlinde-like bridge is not derived internally.
3. Paper VII is not a full SPARC rotation-curve likelihood with galaxy-level distance, inclination, stellar mass-to-light, covariance, gas-systematic, external-field and interpolation-family nuisance treatment.
4. No unique interpolation function is derived from the action.
5. Active38 is not an independent  $a_0$  anchor.

## 14 Conclusions

1. Paper VII v0.3 locks the BPB/MBE acceleration-scale gate at  $a_0^{\text{BPB}} = 1.367856592595 \times 10^{-10} \text{ m s}^{-2}$ .
2. The value is fixed by  $cH_0(1 - s_{\text{surv},0})$ , not fitted to galaxy data.
3. The same value is reproduced by the conditional D2 thermal route through  $T_{\text{eff}} = (1 - s_{\text{surv},0})T_{\text{dS}}$ .
4. The effective BTFR identity follows from  $g_N(r_t) = a_0^{\text{BPB}}$ :  $V_f^4 = GM_b a_0^{\text{BPB}}$ .
5. On the SPARC BTFR sample, the fixed- $a_0$  prediction gives  $n = 118$ , median residual  $-0.0560$  dex and RMS  $0.2955$  dex.
6. A minimal BTFR likelihood supports the normalisation: a free slope-4 offset is not required by BIC.
7. The free-slope model is preferred, so slope/interpolation closure is not claimed.
8. Fixed- $a_0$  RAR interpolation improves raw point-level baryonic SPARC rotation curves with fractional raw- $\chi^2$  improvement  $0.9112$ .
9. Global nuisance profiling gives primary best  $a_0/a_0^{\text{BPB}} = 0.8071$  with fixed penalty fraction  $0.0520$ .
10. The Paper-II active38 subset is retained only as a shape/interpolation cross-check.
11. The final allowed claim is fixed- $a_0$  BTFR/interpolation support.

12. The forbidden claims are full SPARC RC likelihood closure, exact galaxy-by-galaxy MOND closure, a unique interpolation-function derivation, active38 as an independent  $a_0$  anchor, and unconditional action-level MOND derivation.

## A v0.3 source files and reproduction

The v0.3 source package contains:

```
paperVII_BTFR_a0_v0_3.tex
data/paperVII_v0_3_numeric_locks.csv
data/paperVII_v0_3_gate_dashboard.csv
data/paperVII_v0_3_claim_source_lock.csv
scripts/make_paperVII_v0_3_figures.py
figs/fig1_a0_scale_v02.pdf/png
figs/fig2_gate_dashboard_v02.pdf/png
figs/fig3_btfr_residuals_v02.pdf/png
figs/fig4_btfr_likelihood_bic_v02.pdf/png
figs/fig5_raw_rc_improvement_v02.pdf/png
figs/fig6_a0_profile_ratios_v02.pdf/png
release_index/00_MANIFEST_CORPUS.csv
release_index/00_CLAIMS_INDEX.csv
release_index/00_REVIEWER_ROUTE.md
```

The figures can be regenerated with:

```
python scripts/make_paperVII_v0_3_figures.py
```

The full P7-A1–P7-A7 source lock is inherited from the repository outputs:

```
outputs/paperVII/paperVII_P7A1_constants_a0_lock.*
outputs/paperVII/paperVII_P7A2D2_thermal_bridge.*
outputs/paperVII/paperVII_P7A2D2b_D2_source_provenance.*
outputs/paperVII/paperVII_P7A3_BTFR_normalization_test.*
outputs/paperVII/paperVII_P7A4_BTFR_likelihood_gate.*
outputs/paperVII/paperVII_P7A5_interpolation_raw_baryonic_rc.*
outputs/paperVII/paperVII_P7A6_full_rc_likelihood_or_interpolation_family.*
outputs/paperVII/paperVII_P7A6b_active_boundary_audit.*
outputs/paperVII/paperVII_P7A7_manuscript_lock.*
```

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